Spectral Score v0.2 — absorbs Elie's 3 bulk radial towers (Toy 3627), Grace Pair α intermediate Casimirs {0,4,6}, bulk-color v0.5 Toeplitz 8 = 3+3+2 count match, Quasi-Eigentone v0.2 Resolution B, and Mersenne-Hall connection. The V0 narrative spine is richer: each sector now has a complete radial tower; decay arrows have channel labels; the gluon row has a candidate 8-generator structure.

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 10:45 EDT (date-verified)

Spectral Score v0.2 — Saturday absorptions

0. What v0.2 absorbs

v0.1 (Saturday earlier, before timestamp recalibration) gave the structural backbone. v0.2 absorbs the data that landed since:

1. Elie Toy 3627 — three bulk radial towers with closed-form Casimirs.
2. Grace 1.5 Pair α — spinor^3 intermediate Casimirs {0, 4, 6} for generations.
3. Bulk-color v0.5 — Toeplitz SU(3) count match 8 = 3 + 3 + 2.
4. Quasi-Eigentone v0.2 — Resolution B (T190 exponent = gauge-mediator C_2).
5. Mersenne-Hall connection — Hall algebra structure constants are Mersenne integers.

The V0 movie narrative spine is now richer + more concretely tabulated.

1. Sector radial towers (Elie Toy 3627)

Each sector now has a complete radial tower with closed-form Casimirs:

Sector	Tower	Closed form	First 5 Casimirs
Scalar (Higgs)	trivial V_(0,0)	—	0
Vector (gauge bosons radial)	V_(k,0)	k(k+3)	0, 4, 10, 18, 28
Adjoint (gauge sector radial)	V_(k,k) = V_(0,2k)	2k(k+2)	0, 6, 16, 30, 48
Spinor (lepton matter radial)	V_(k+1/2, k+1/2) = V_(0,2k+1)	(k+1/2)(2k+5)	5/2, 21/2, 45/2, 77/2, 117/2

Movie visualization: each sector is a vertical "column" with the radial tower's Casimirs spaced on it. Substrate-primary anchors mark specific levels: - Vector tower's k=1 = Casimir 4 = rank^2 (photon-like anchor). - Adjoint tower's k=1 = Casimir 6 = C_2 (gauge-sector anchor). - Spinor tower's k=0 = Casimir 5/2 = ρ_1 (lepton-anchor / natural mass unit).

2. Generation channel structure (Grace Pair α + Quasi-Eigentone v0.2 Resolution B)

The 3 lepton generations correspond to the 3 spinor^3 channels via the BOSONIC INTERMEDIATE Casimirs:

Generation	Particle	Channel intermediate	Intermediate Casimir	Resolution B mapping
1	e, ν_e	trivial-channel	0	TRUE eigentone
2	μ, ν_μ	adjoint-channel	C_2 = 6	QUASI; mediator = adjoint of so(5)
3	τ, ν_τ	vector-channel	rank^2 = 4	QUASI; mediator = vector

(Resolution B mapping per Quasi-Eigentone v0.2; alternative Resolution A would reverse gens 2 and 3.)

Movie visualization: each generation's decay arrow goes through the bosonic-intermediate K-type: - $\mu \to$ e: arrow labeled "via adjoint" (mediator V_(1,1)) - $\tau \to$ e: arrow labeled "via vector" (mediator V_(1,0)) - $\tau \to \mu$: composite arrow

3. Gluon octet candidate (bulk-color v0.5)

The 8 gluons of SU(3)_C now have a CANDIDATE substrate-natural decomposition:

8 = 3 (SO(3)-vector Toeplitz T_a) + 3 ([T_a, T_b] Hankel commutators) + 2 (K rank-2 Cartan)

Movie visualization: the gluon row is split into THREE sub-rows: - 3 SO(3)-vector Toeplitz operators T_a (a = 1, 2, 3) — color labels. - 3 commutator-derived operators [T_a, T_b] — non-abelian Hankel content. - 2 K-Cartan operators — rank-2 Cartan of K = SO(5)×SO(2).

This is a candidate match (8 numerically), not a closed derivation. Movie can SHOW the 3+3+2 decomposition as a structural sub-grouping of the gluon row.

4. Substrate-primary anchors (Mersenne-Hall connection)

Each substrate primary that appears in K-type Casimirs is naturally a low Mersenne integer or product: - 3 = N_c = M_2 (color, dual Coxeter) - 7 = g = M_3 (signature) - 15 = N_c·n_C = M_4 (Sym^2(5) dim) - 21 = N_c·g (dim so(5,2)) - 127 = M_7 (anchors N_max = 128 + 9)

Movie visualization: the substrate-primary anchors on the spectral score can carry small Mersenne-tower labels next to them — visualizing the Mersenne backbone of the substrate's structural integers.

5. T190 closed-form on the score (Quasi-Eigentone v0.2 Resolution B)

m_μ/m_e = (24/π^2)^{C_2} = (rank^3·N_c/π^2)^{C_2}

Visualized on the score: - e $\to \mu$ arrow with label "(rank^3·N_c/π^2)^{C_2}". - The exponent C_2 = 6 visually connects to the adjoint K-type V_(1,1) row, showing the gauge-mediator interpretation.

6. Updated V0 movie narrative (richer)

Frame 1: Bare substrate D_IV^5 — empty Casimir spectrum, the bounded symmetric domain. Frame 2: K-type lattice appears — the substrate's spectral lattice lights up. Substrate-primary anchors (0, 5/2, 4, 6, 21/2, …) glow as Mersenne-Hall integers. Frame 3: SM particles populate — each particle drops onto its K-type slot. Channel labels appear: - Higgs at V_(0,0) Casimir 0 (trivial channel). - Photon at V_(1,0) Casimir 4 (vector channel). - Gauge W, Z at V_(1,1) Casimir 6 (adjoint channel). - Leptons at V_(1/2,1/2) Casimir 5/2 = ρ_1 (natural mass unit; matter sets the unit). - Proton at bulk V_(1,1) k=6 Casimir 6 = C_2. Frame 4: Bulk radial towers (Elie 3627) extend upward in each sector — visualizing the available radial excitations. Frame 5: Quasi-eigentones (μ, τ, n, W, Z, H, hadrons) pulse and emit decay arrows along Green coproduct paths with channel labels. Frame 6: Toeplitz-derived gluon octet structure appears — 3 + 3 + 2 decomposition of the gluon adjoint highlighted as a substrate-natural sub-structure. Frame 7: Five-Absence gaps remain dark — no 4th generation slot, no SUSY doublings, no GUT leptoquarks, no Majorana, no monopoles. Frame 8: T190 closed form $(24/\pi^2)$^{C_2} animates from e to μ; T2003, T187 similarly — the substrate's mass-ratio closed forms. Frame 9: Closing — the full SM as the substrate's spectral signature, with all substrate primaries labeled.

7. Honest scope + tier

RIGOROUS (today's team work): - Elie bulk radial towers (Toy 3627, closed forms verified). - Grace intermediate Casimirs {0, 4, 6} (Pair α rep-theoretic). - Toeplitz count match 8 = 3 + 3 + 2 (numerical alignment from Lyra v0.5). - Mersenne-Hall integer connection.

SYNTHESIS (v0.2): incorporates all four pieces into a richer V0 movie narrative; channel-labeled decay arrows; gluon-octet sub-structure visualization; substrate-primary anchors visible.

OPEN (multi-week): explicit construction of the V0 movie artifact (Grace + Elie + visual rendering); the Toeplitz 8-count closure verification.

Cal #27 / honesty: v0.2 is SYNTHESIS for visualization. The Toeplitz 8-count match is a numerical alignment, not derivation. Resolution B for gen-channel mapping is the v0.2 working hypothesis. NOT a derivation; richer narrative for V0 artifact build.

Routed: → Grace/Elie: V0 artifact build using this richer v0.2 spec (sector radial towers + channel labels + gluon sub-structure + Mersenne anchors). → me: continuing v0.2 sweep — next Vol 16 §3 v0.2.

— Lyra, Spectral Score v0.2. Absorbs Saturday's team data into the V0 narrative spine. Per-sector RADIAL TOWERS (Elie Toy 3627), GENERATION CHANNEL labels (Grace Pair α + Resolution B), GLUON OCTET candidate sub-structure 3+3+2 (bulk-color v0.5), MERSENNE-HALL substrate-primary anchors. 9-frame movie narrative refined. Ready for V0 artifact build.